THE HOM COMPLEX FROM A BOUNDED ABOVE PROJECTIVE COMPLEX REPRESENTING A PERFECT OBJECT TO A K-FLAT COMPLEX IS AGAIN K-FLAT

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The author refers to 'Lemma ... and its proof' when identifying the Hom complex with the total tensor complex; that pointer is recorded as a non-load-bearing navigation reference. The eight references inside the counted proof span are recorded as non-load-bearing navigation pointers, mixing cross-chapter tags with same-chapter label lines.

Allowed standard prerequisites (editorial).

- K-flat complexes and their homotopy invariance under the derived tensor product. used at proof lines 28730-28733 and 28743-28744. Is the property being established and the invariance that permits the replacement by F^bullet.
- Representation of a perfect object by a finite complex of finite projective modules, and homotopy equivalence of complexes of projectives that are isomorphic in the derived category. used at proof lines 28721-28727. Supplies the finite complex F^bullet and the homotopy equivalence.
- The dual complex E^bullet of a finite complex of finite projectives and the identification of Hom with a total tensor complex. invoked at proof lines 28734-28742. Rewrites the Hom complex in the form to which the tensor-product K-flatness applies.

Source reference. Original definition 14185: sources/stacks-more-algebra/pinned-original-source.tex, source lines 14185–14185. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original definition 18871: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18871–18871. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 064B: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7860–7860. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 18708: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18708–18708. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 14104: sources/stacks-more-algebra/pinned-original-source.tex, source lines 14104–14104. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 14284: sources/stacks-more-algebra/pinned-original-source.tex, source lines 14284–14284. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 19176: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19176–19176. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 14224: sources/stacks-more-algebra/pinned-original-source.tex, source lines 14224–14224. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a ring. Let $P^\bullet$ be a bounded above complex of projective R-modules. Let $K^\bullet$ be a K-flat complex of R-modules. If $P^\bullet$ is a perfect object of $D(R)$, then $\mathrm{Hom}^\bullet(P^\bullet, K^\bullet)$ is K-flat and represents $R\mathrm{Hom}_R(P^\bullet, K^\bullet)$.*

Proof. The last statement is Lemma 18708. Since $P^\bullet$ represents a perfect object, there exists a finite complex of finite projective R-modules $F^\bullet$ such that $P^\bullet$ and $F^\bullet$ are isomorphic in $D(R)$, see Definition 18871. Then $P^\bullet$ and $F^\bullet$ are homotopy equivalent, see Derived Categories, Lemma 064B. Then $\mathrm{Hom}^\bullet(P^\bullet, K^\bullet)$ and $\mathrm{Hom}^\bullet(F^\bullet, K^\bullet)$ are homotopy equivalent. Hence the first is K-flat if and only if the second is (follows from Definition 14185 and Lemma 14104). It is clear that

$$\mathrm{Hom}^\bullet(F^\bullet, K^\bullet) = \mathrm{Tot}(E^\bullet \otimes_R K^\bullet)$$

where $E^\bullet$ is the dual complex to $F^\bullet$ with terms $E^n = \mathrm{Hom}_R(F^{-n}, R)$, see Lemma 19176 and its proof. Since $E^\bullet$ is a bounded complex of projectives we find that it is K-flat by Lemma 14284. Then we conclude by Lemma 14224. □